

MASTER'S COMPREHENSIVE EXAMINATION

Part II Applied (2 hours) March 28th, 2015

Instructions: Attempt any three out of the following four questions on both pages. Mention which three you would like to have graded. Open book and open notes.

1. A supermarket buyer seeks to compare the proportion of cracked eggs in a shipment from two suppliers A and B. The buyer's research hypothesis is that B has a greater proportion of cracked eggs. The eggs come 12 to a carton, and the buyer picks a simple random sample of 15 cartons from each supplier. The following are the results:

Number of cracked eggs		Number of cartons with X cracked eggs	
In Carton	X	Supplier A	Supplier B
0		13	10
1		2	0
2		0	0
3		0	0
4		<u>0</u>	<u>5</u>
Total cartons		15	15
Total Eggs		180	180
Total cracked eggs		2	20

For each of the following tests state whether it is appropriate or not **to test H_0 of no difference in proportions of cracked eggs**, versus H_a that B has a greater proportion of cracked eggs. (circle Yes or No) If No, explain why:

a) A t-test for the difference of two proportions, based on independent samples of $n_1 = n_2 = 180$ eggs. Yes or No. _____

b) Wilcoxon-rank sum test for the difference of the mean number of cracked eggs per carton from supplier A vs supplier B Yes or No. _____

c) McNemar's test for the difference of paired proportions.
Yes or No. _____

d) Fisher's Exact test for the following 2x2 table.

	Supplier A	Supplier B
All good eggs	13	10
Some cracked eggs	2	5

Yes or No. _____

e) A t-test for the difference of the mean number of cracked eggs per carton from supplier A vs supplier B Yes or No. _____

2. The amount of rent a snack salesman for a stand at “sleaze-side beach” on any given day during April 2015 is based on the weather that day as follows

\$25 for the day if it rains

\$100 for the day if it is cloudy but does not rain

\$175 for the day if it is sunny

He has data for April from the past 10 years and observed that; it rains 25% of the days, is cloudy not raining 60% of the days and is sunny 15% of the days. You can take these numbers to be constant and representative of what will happen in 2015.

A. For any given day, what is the expected amount he pays and the variance of this?

B. Assume that the weather on each of the 30 days of April is independent from that of the other days and estimate the probability the salesman will have to pay more than \$3,000 in rent for April?

3. Dr. Bill Tirecheck wants to compare two adjacent facilities Tombo’s Tire Rack and Giso’s Tire Shop on maintaining air pressure in the tires over time. For five consecutive months a new tire is sent to each place on the first day of that month. The air pressure of that tire is measured at 10 AM on the first day of the of the month and then again at 10 AM on the last day of the month. There are strong random calendar effects on measured tire pressure including temperature and humidity. These factors randomly differ day to day, but as both facilities are next to each other, these factors are the same for both facilities on the same given day. The following results were seen.

TOMBO’S TIRE RACK			GISO’S TIRE SHOP		
Month New Tire Sent	Tire Pressure on Day of Month		Month New Tire Sent	Tire Pressure on Day of Month	
	1 st Day	Last Day		1 st Day	Last Day
Jan 2016	14	10	Jan 2016	13	10
Feb 2016	20	15	Feb 2016	16	13
Mar 2016	20	16	Mar 2016	19	19
Apr 2016	22	17	Apr 2016	20	17
May 2016	27	25	May 2016	27	25

To answer the questions below, you can assume that the observations are normally distributed with constant variance and except as noted above are independent

A. Dr. Tirecheck believes that tires sent to “Tombo’s Tire Rack” systematically had higher pressure to begin with than did those sent to “Giso’s Tire Shop”. State H_0 and H_a and test at $\alpha=0.05$.

B. Dr. Tirecheck also believes that there was a difference in change in tire pressure form the first to last day between “Tombo’s Tire Rack” and “Giso’s Tire Shop”. State H_0 and H_a and test at $\alpha=0.05$.

4. A researcher wants to compare the effectiveness of three anti-wrinkle creams as used for different age groups. The response variable is a numerical measure on a ratio scale of the reduction of wrinkles. 6 doctors participate in the study. 2 doctors are assigned at random to each cream. Each doctor has 4 participating patients aged 40-45, 4 participating patients aged 50-55, and 4 participating patients aged 60-65. The researcher views the doctors as a random sample of doctors, and the patients as a random sample of their patients within those age groups. The data layout is as follows.

	Cream 1		Cream 2		Cream 3	
	Doctor 1	Doctor 2	Doctor 3	Doctor 4	Doctor 5	Doctor 6
40-45	4	4	4	4	4	4
50-55	4	4	4	4	4	4
60-65	4	4	4	4	4	4

Write out the ANOVA table showing source of variability, degrees of freedom, Expected Mean Squares, and show which F ratios to take.